

Sante-Price Index Android App - Full Kotlin Source Code

This document contains complete Android Studio Kotlin source code for the Sante-Price Index (Micro-Finance) project.

MainActivity.kt

```
package com.example.santepriceindex

import android.content.Intent
import android.os.Bundle
import android.widget.*
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    lateinit var txtPrices: TextView
    lateinit var edtCost: EditText
    lateinit var txtResult: TextView

    var onionPrice = 25.0

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        txtPrices = findViewById(R.id.txtPrices)
        edtCost = findViewById(R.id.edtCost)
        txtResult = findViewById(R.id.txtResult)

        val btnCalculate = findViewById<Button>(R.id.btnCalculate)
        val btnBoard = findViewById<Button>(R.id.btnBoard)
        val btnTrend = findViewById<Button>(R.id.btnTrend)

        txtPrices.text =
            "Today's Onion Mandi Price: ■$onionPrice"

        btnCalculate.setOnClickListener {

            val transport =
                edtCost.text.toString().toDoubleOrNull() ?: 0.0

            val profitMargin = 10
            val wasteCost = 5

            val finalPrice =
                onionPrice + transport + wasteCost + profitMargin

            txtResult.text =
                "Recommended Selling Price: ■$finalPrice per Kg"
        }

        btnBoard.setOnClickListener {

            val intent =
                Intent(this, PriceBoardActivity::class.java)

            intent.putExtra("price", txtResult.text.toString())

            startActivity(intent)
        }

        btnTrend.setOnClickListener {
            startActivity(Intent(this, TrendsActivity::class.java))
        }
    }
}
```

PriceBoardActivity.kt

```
package com.example.santepriceindex
```

```

import android.os.Bundle
import android.widget.TextView
import androidx.appcompat.app.AppCompatActivity

class PriceBoardActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_price_board)

        val txtBoard = findViewById<TextView>(R.id.txtBoard)

        val price =
            intent.getStringExtra("price")

        txtBoard.text = price
    }
}

```

TrendsActivity.kt

```

package com.example.santepriceindex

import android.os.Bundle
import android.widget.TextView
import androidx.appcompat.app.AppCompatActivity

class TrendsActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_trends)

        val txtTrend = findViewById<TextView>(R.id.txtTrend)

        txtTrend.text =
            "Market Trend:\n\nOnion -> Rising\nTomato -> Falling\nPotato -> Stable"
    }
}

```

activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>

<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:orientation="vertical"
        android:padding="16dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content">

        <TextView
            android:text="Sante Price Index"
            android:textSize="28sp"
            android:textStyle="bold"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"/>

        <TextView
            android:id="@+id/txtPrices"
            android:textSize="20sp"
            android:paddingTop="20dp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"/>

        <EditText
            android:id="@+id/edtCost"
            android:hint="Transport Cost"
            android:inputType="numberDecimal"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>
    </LinearLayout>
</ScrollView>

```

```

        <Button
            android:id="@+id/btnCalculate"
            android:text="Calculate Selling Price"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>

        <TextView
            android:id="@+id/txtResult"
            android:textSize="22sp"
            android:textStyle="bold"
            android:paddingTop="20dp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"/>

        <Button
            android:id="@+id/btnBoard"
            android:text="Open Price Board"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>

        <Button
            android:id="@+id/btnTrend"
            android:text="View Trends"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>

    </LinearLayout>

</ScrollView>

```

activity_price_board.xml

```

<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:background="#000000">

    <TextView
        android:id="@+id/txtBoard"
        android:textColor="#FFFF00"
        android:textSize="36sp"
        android:textStyle="bold"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

</LinearLayout>

```

activity_trends.xml

```

<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:gravity="center"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:id="@+id/txtTrend"
        android:textSize="24sp"
        android:textStyle="bold"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

</LinearLayout>

```

AndroidManifest.xml

```

<?xml version="1.0" encoding="utf-8"?>

```

```
<manifest xmlns:android="http://schemas.android.com/apk/res/android">

    <application
        android:allowBackup="true"
        android:label="Sante Price Index"
        android:supportRtl="true"
        android:theme="@style/Theme.AppCompat.Light.DarkActionBar">

        <activity android:name=".TrendsActivity"/>
        <activity android:name=".PriceBoardActivity"/>

        <activity
            android:name=".MainActivity"
            android:exported="true">

            <intent-filter>
                <action android:name="android.intent.action.MAIN"/>
                <category android:name="android.intent.category.LAUNCHER"/>
            </intent-filter>

        </activity>

    </application>
</manifest>
```

Firestore Database Structure

```
{
  "prices": {
    "Onion": 25,
    "Tomato": 30,
    "Potato": 20
  }
}
```